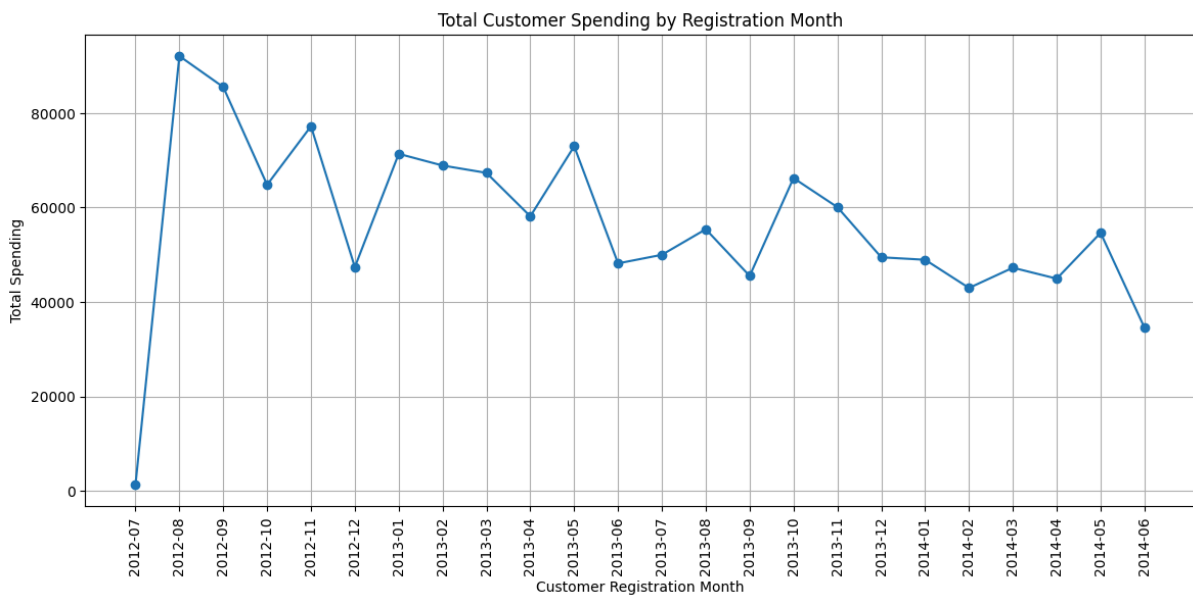
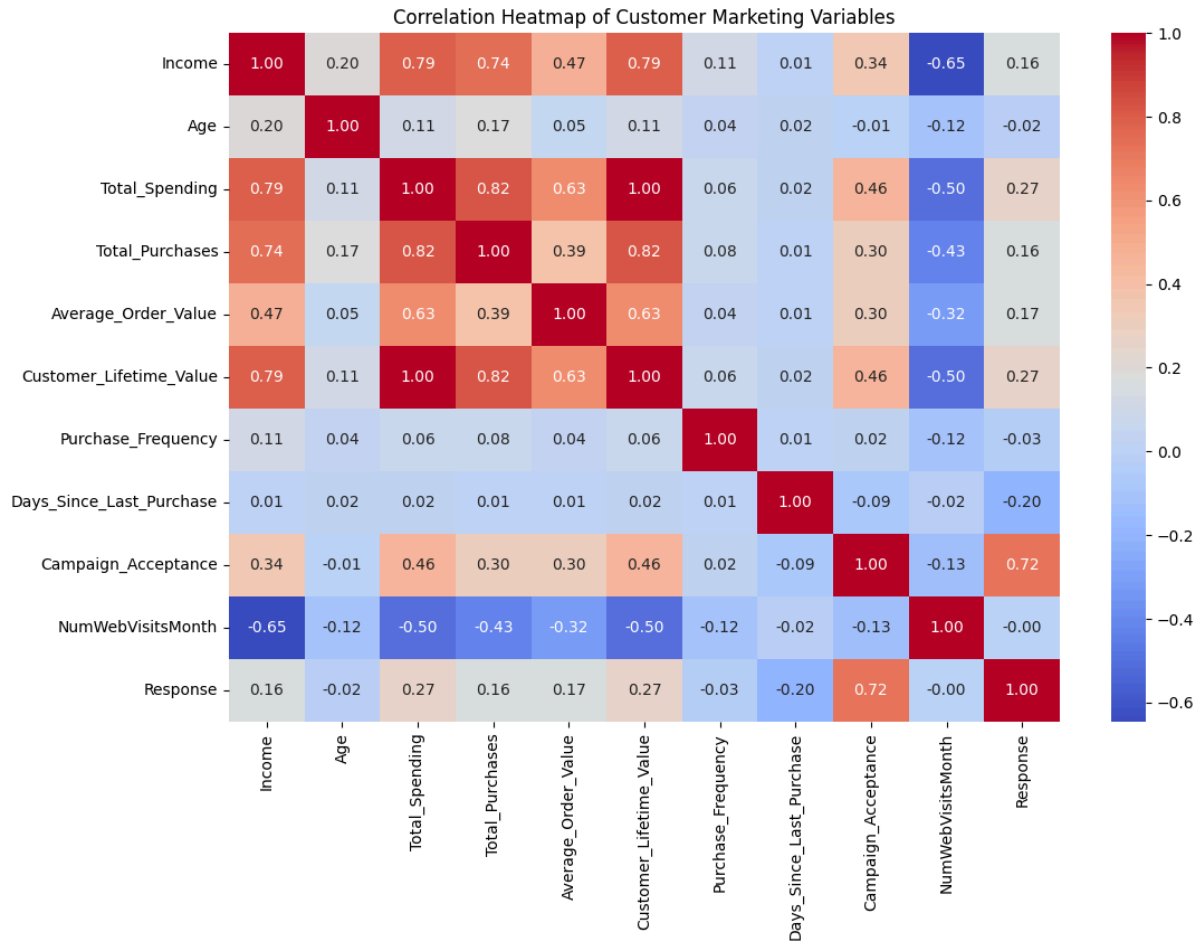
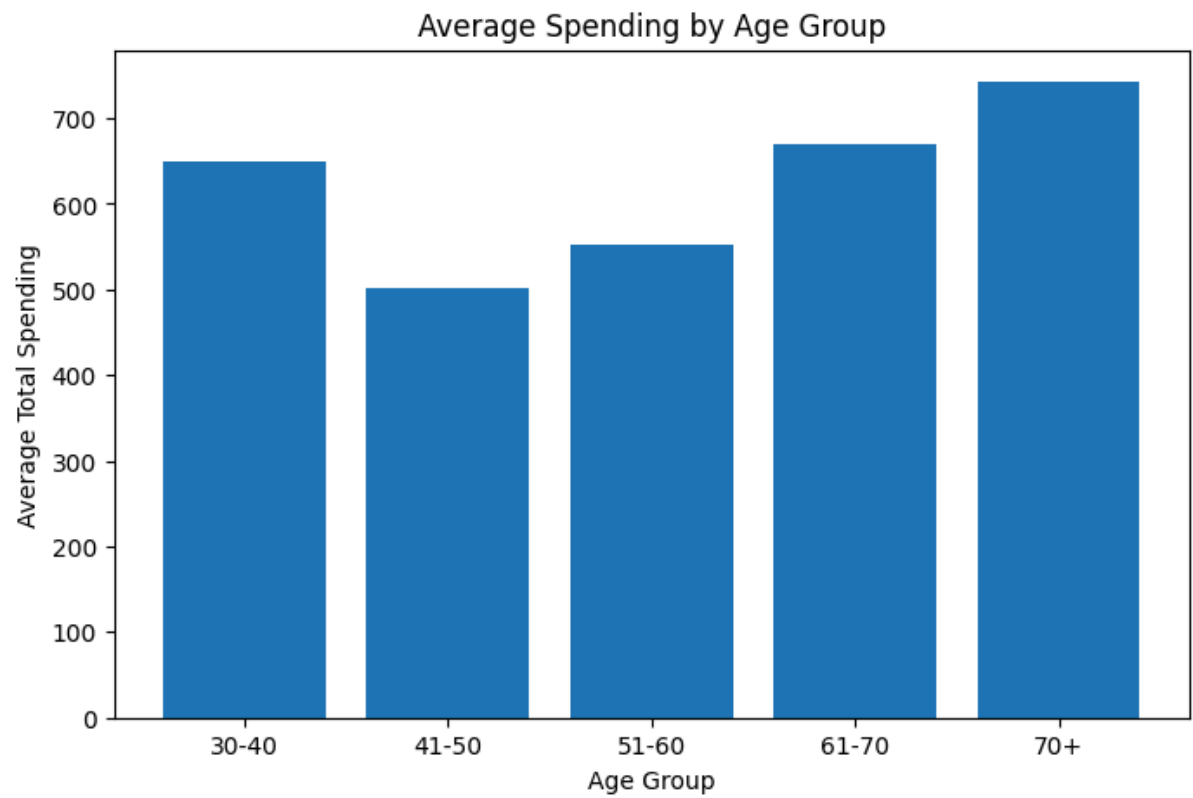
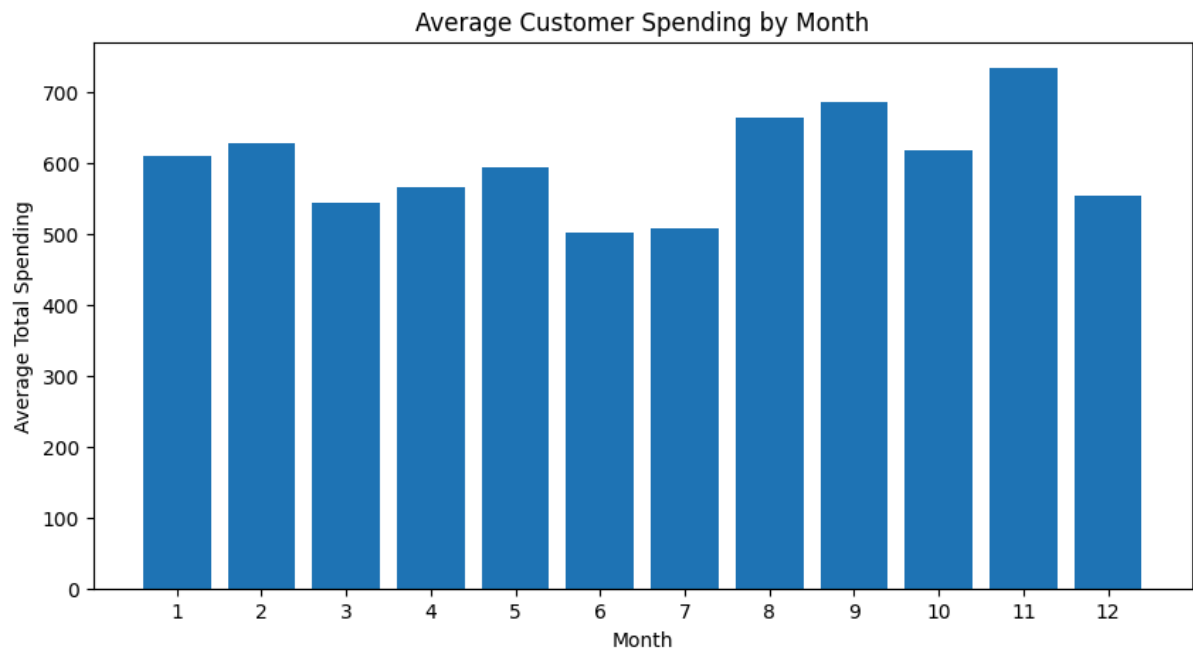
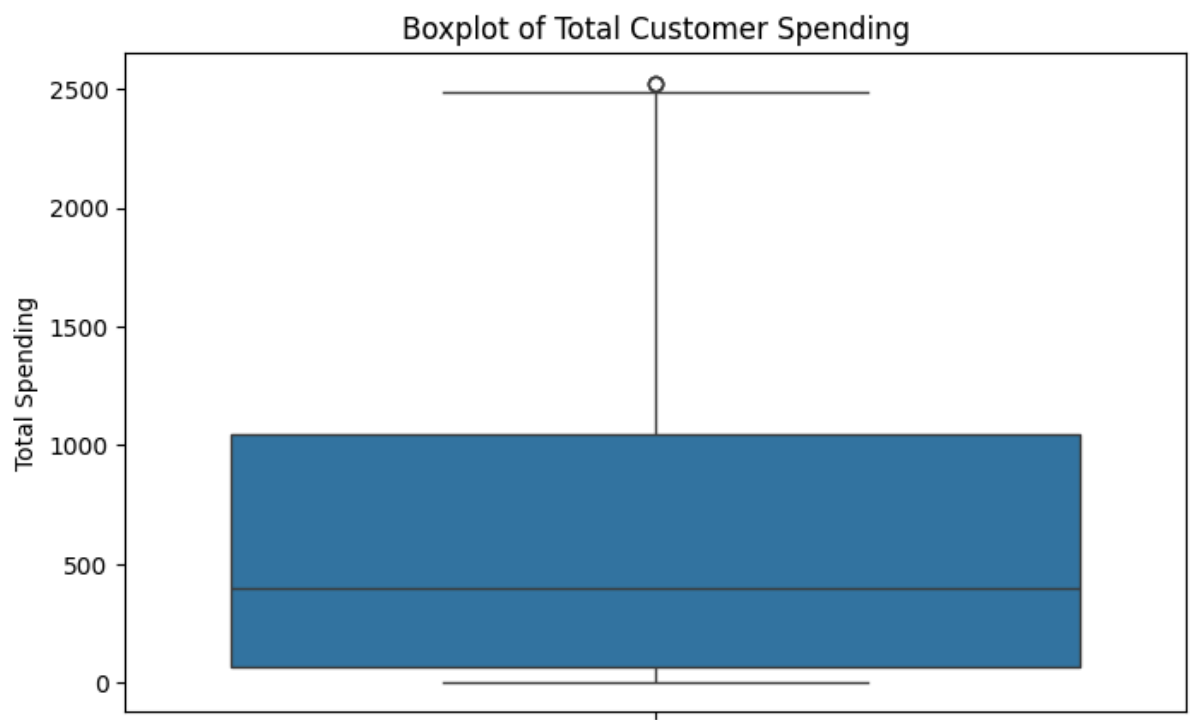
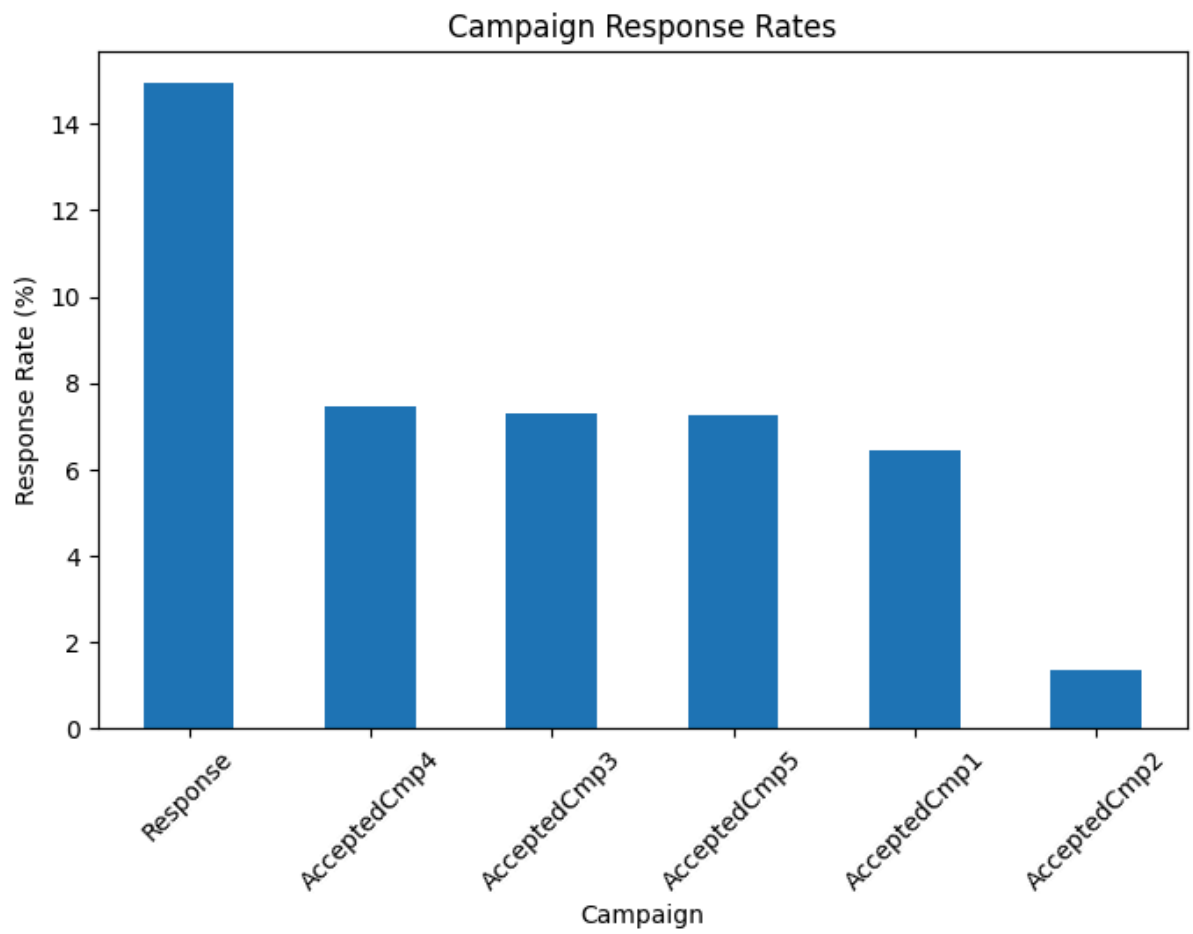


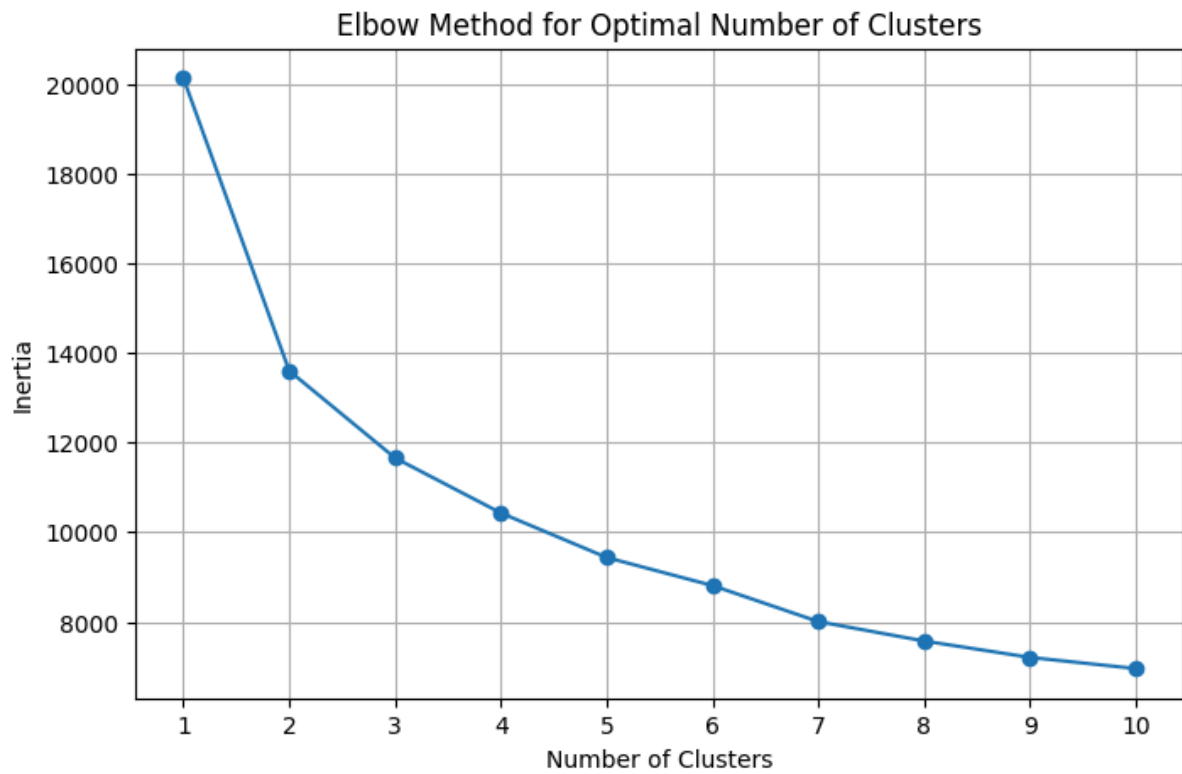
SCREENSHOTS

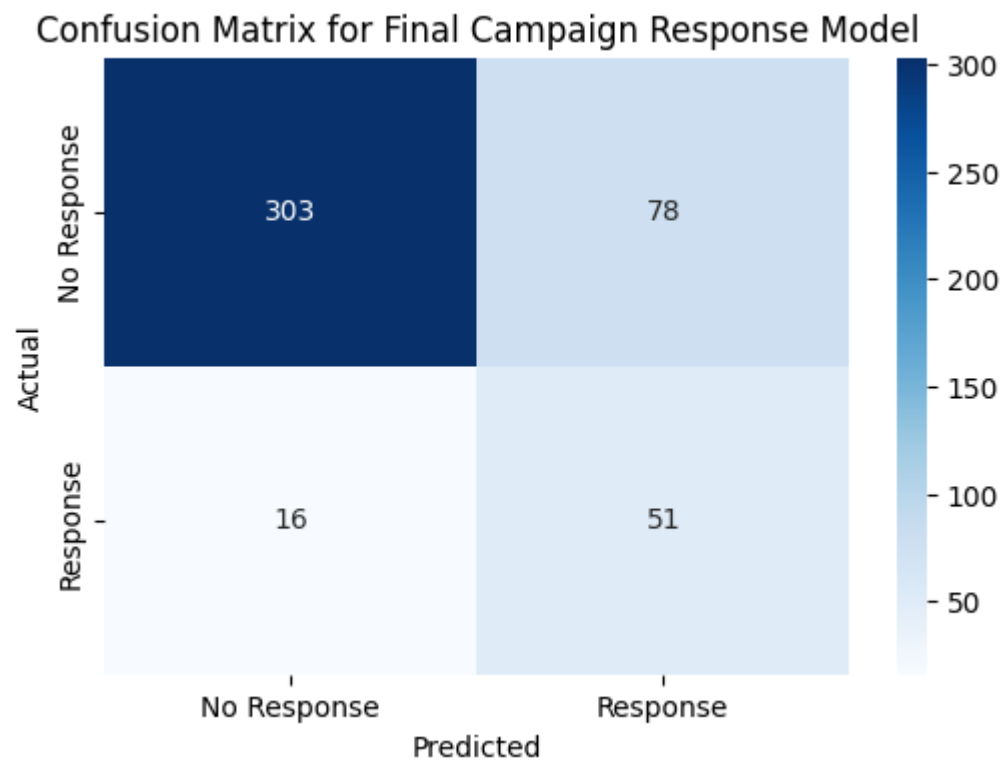
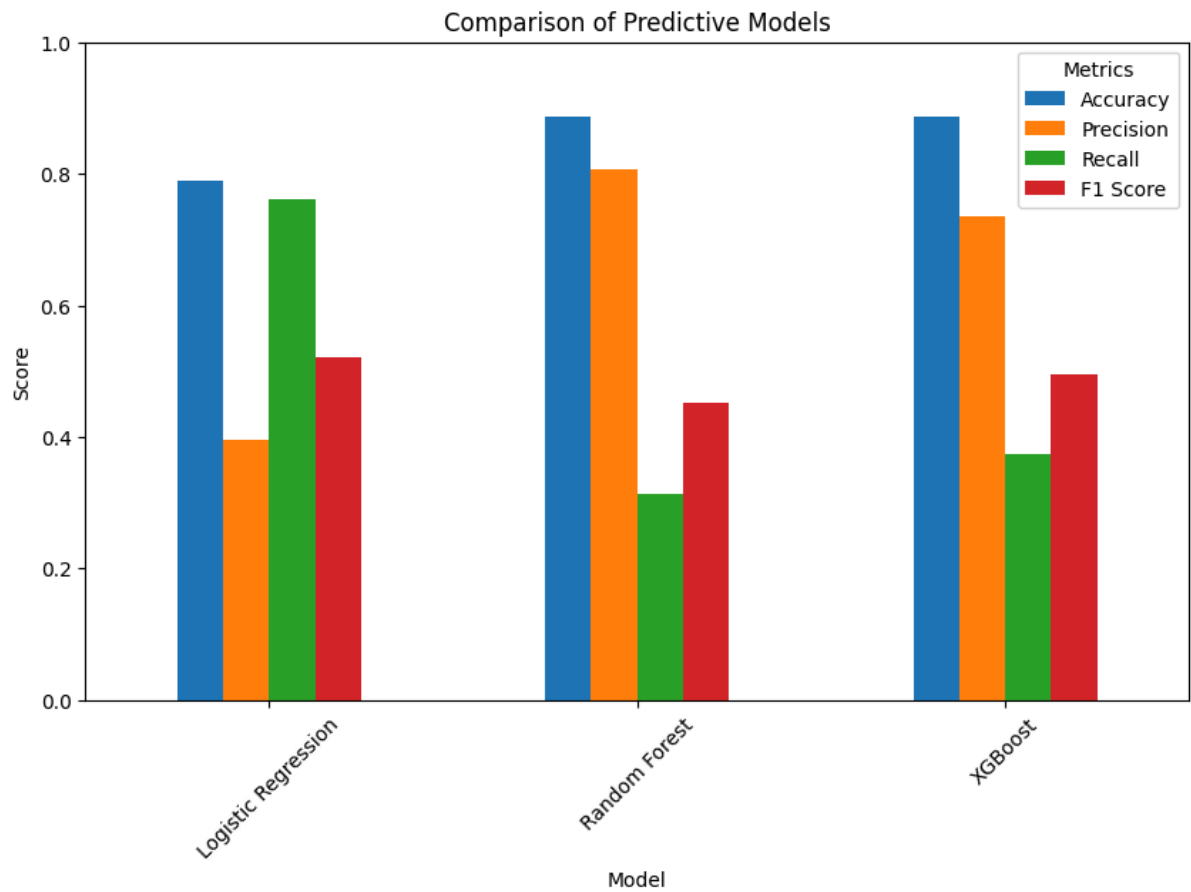
from the ipyb code notebook

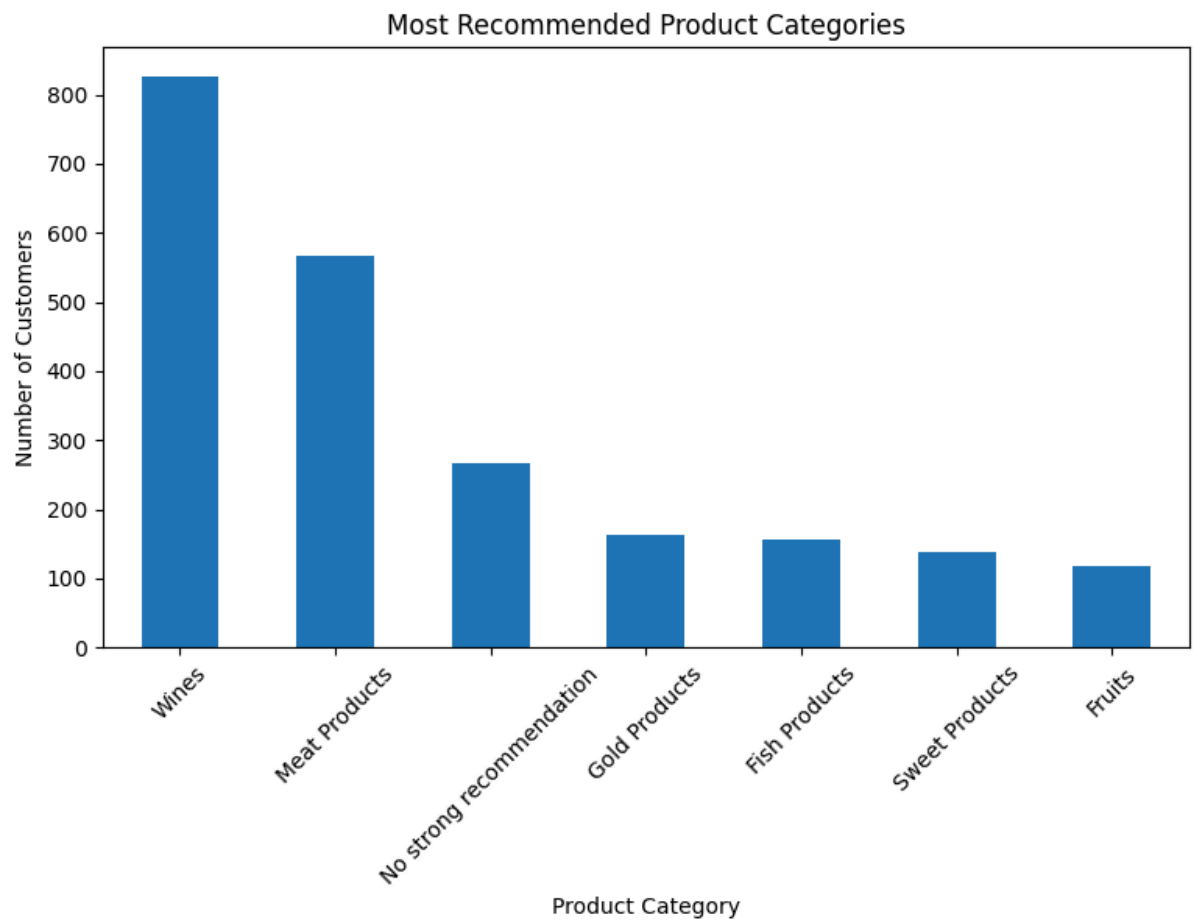
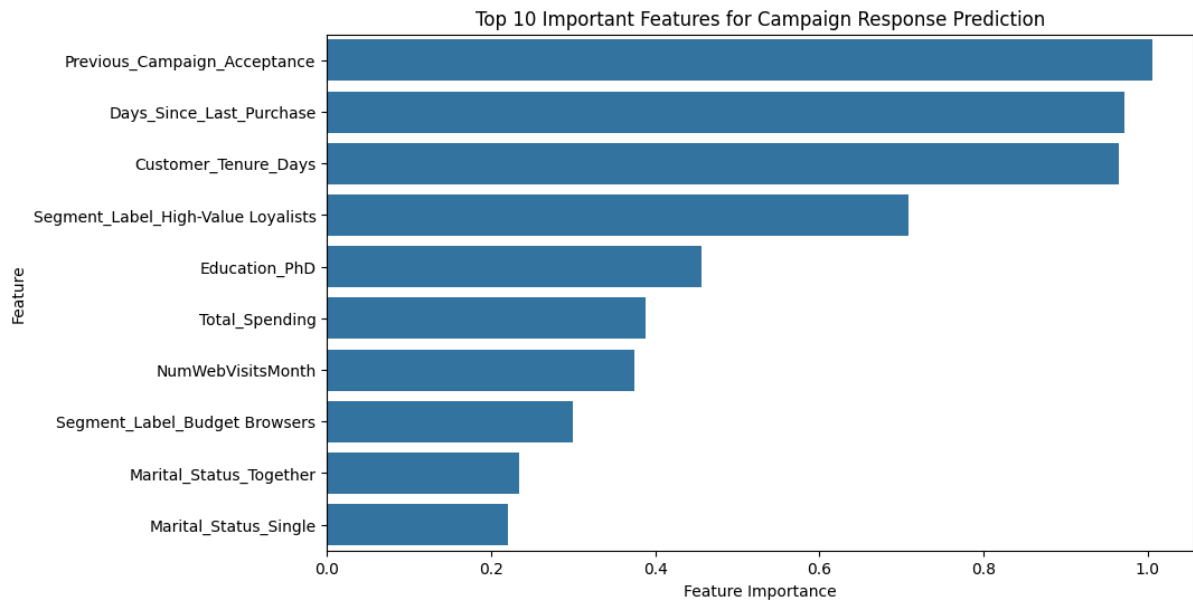


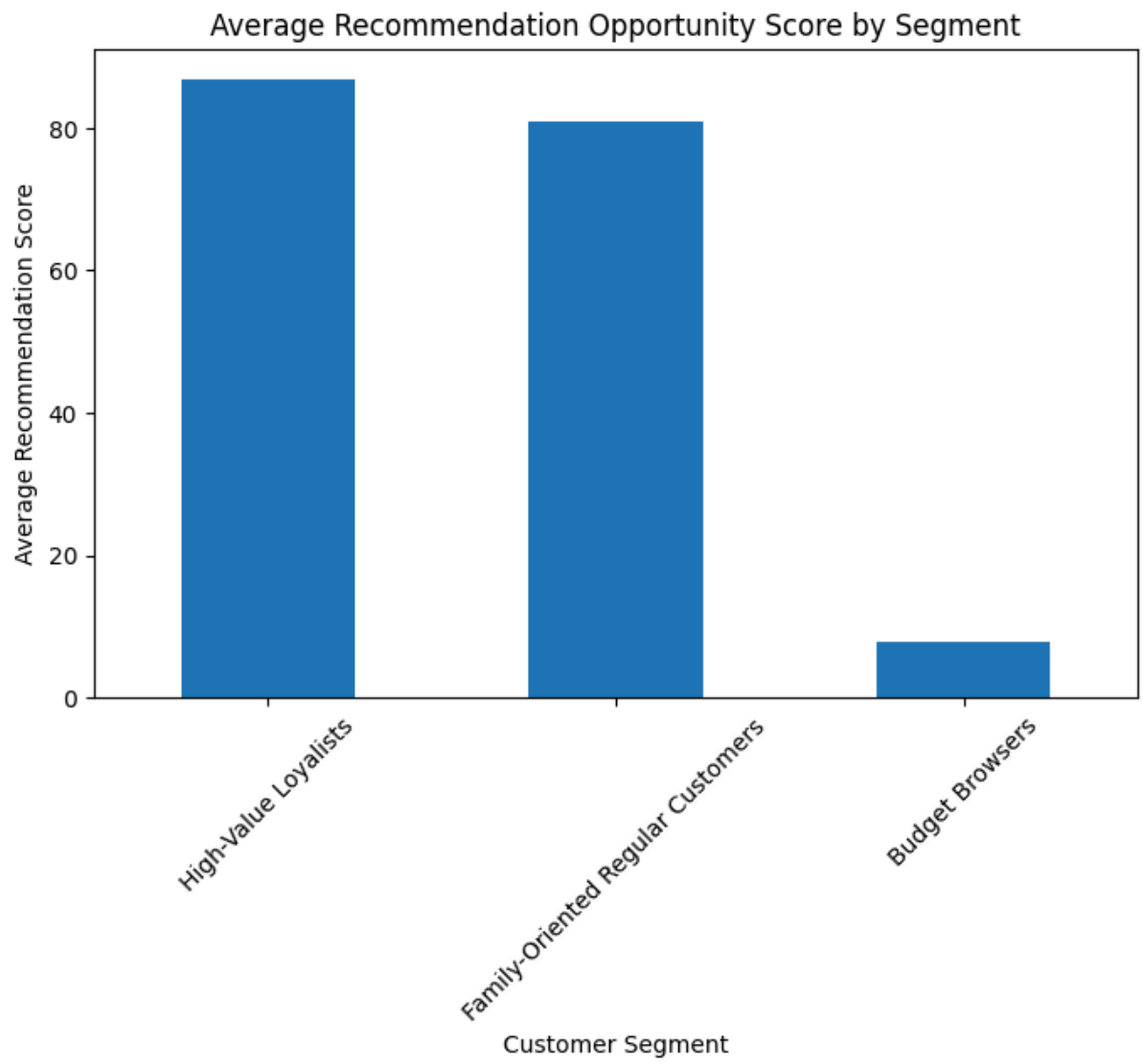


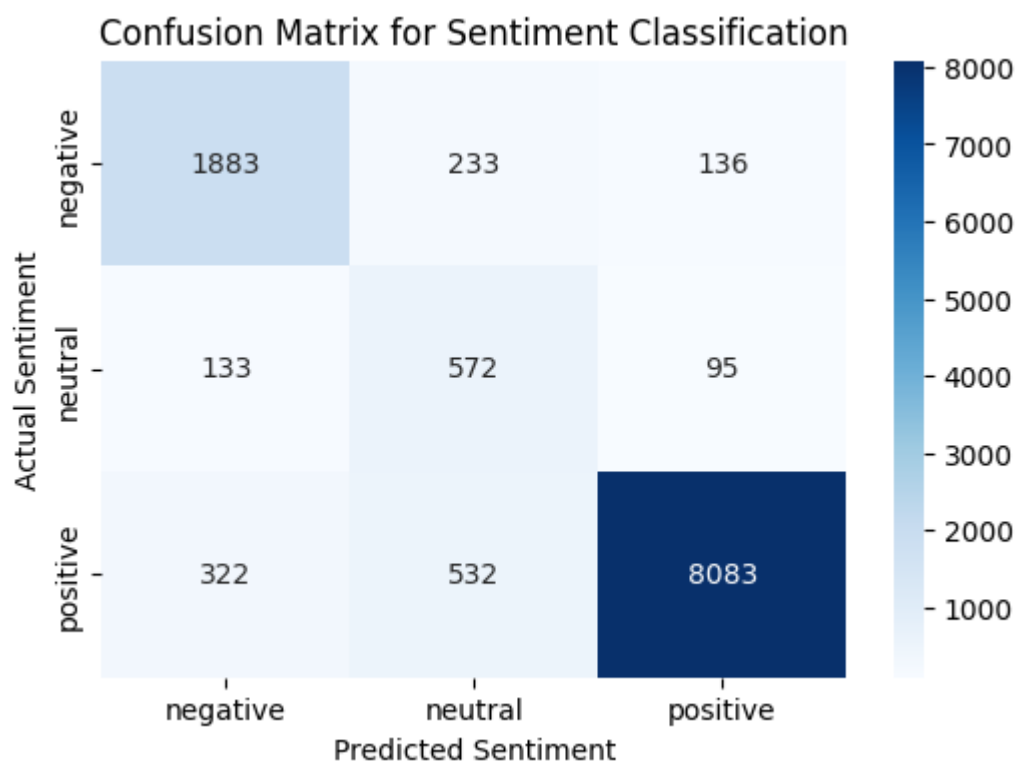
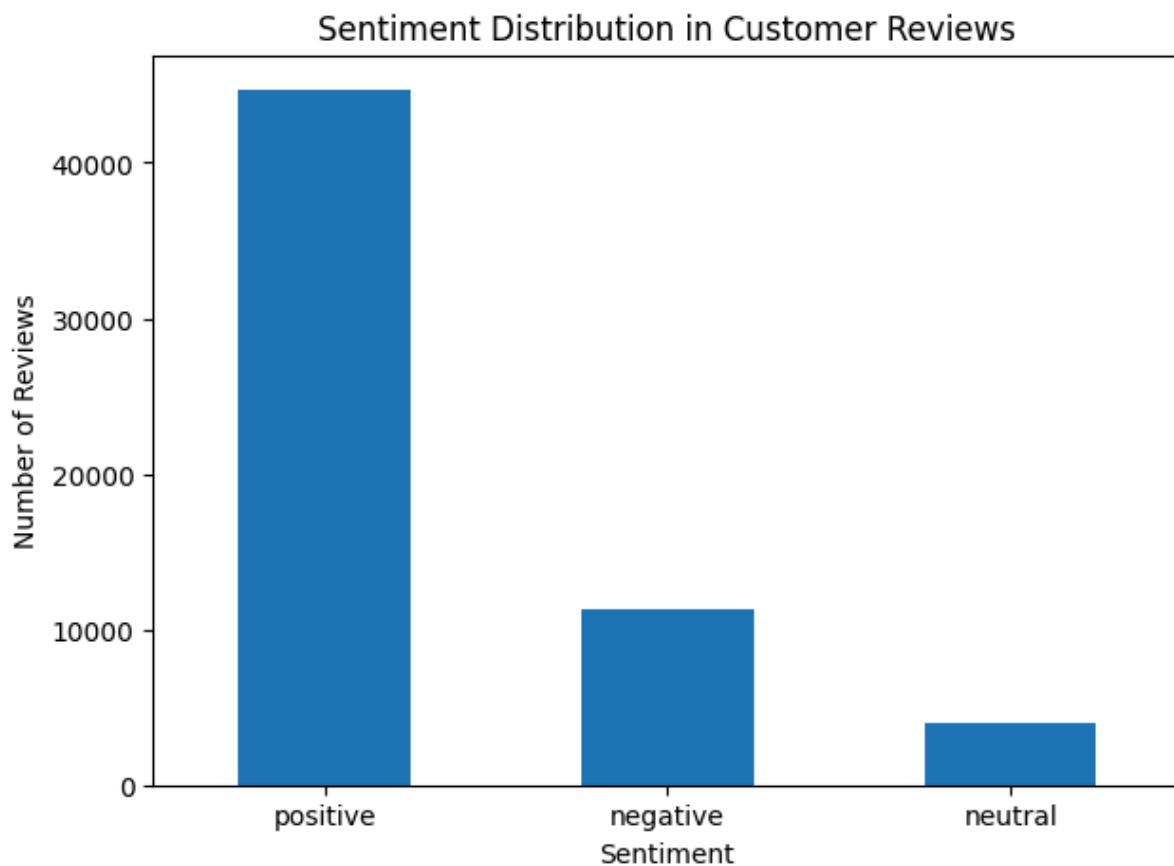




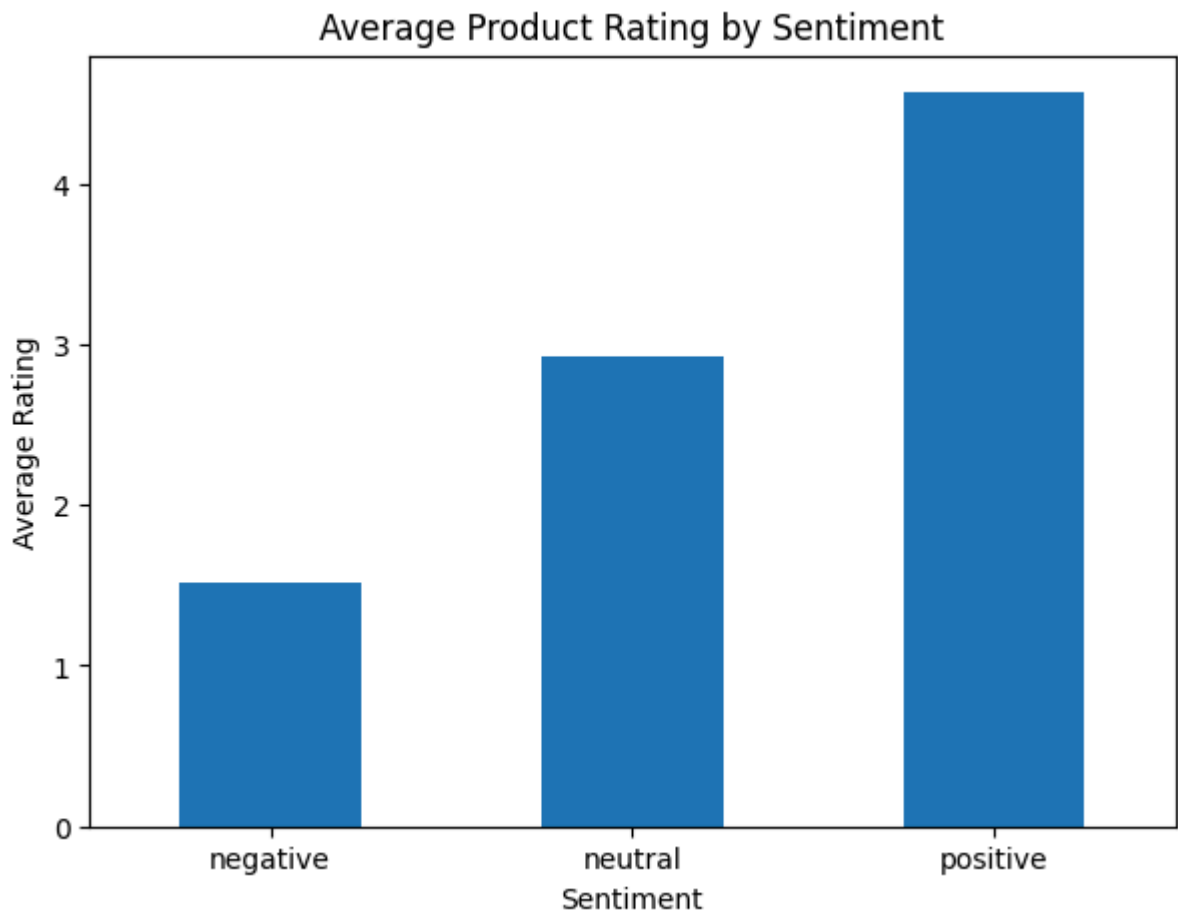
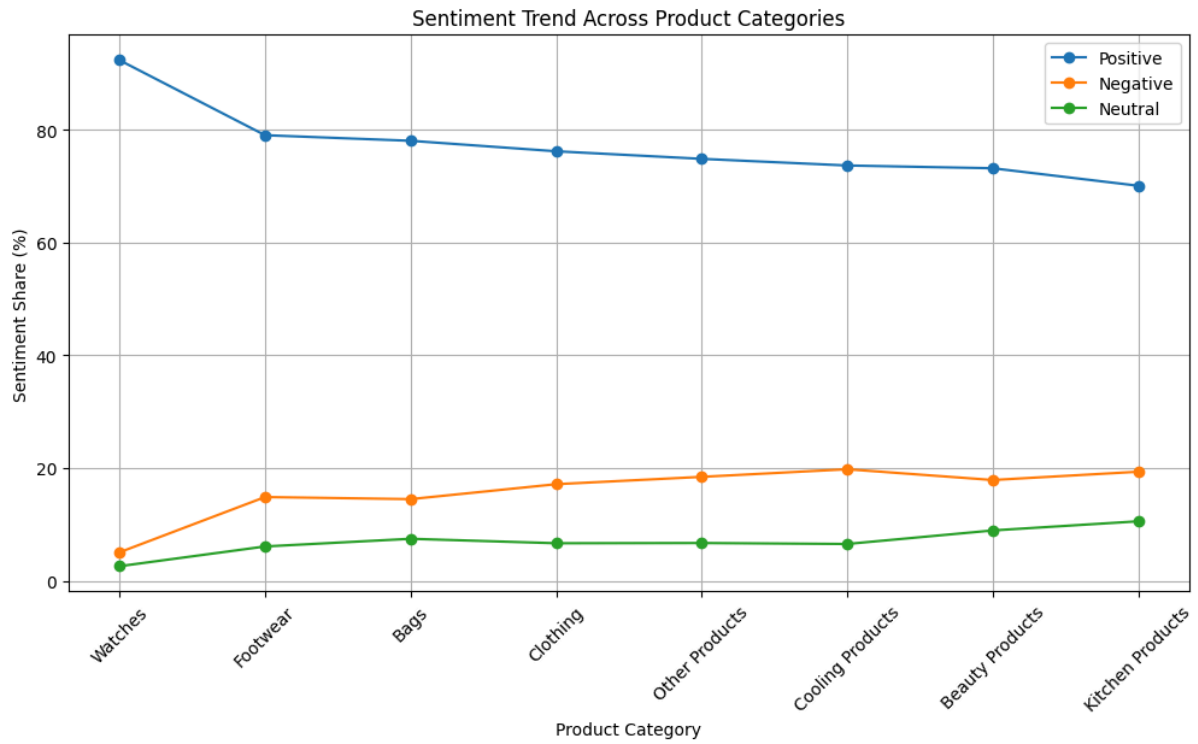




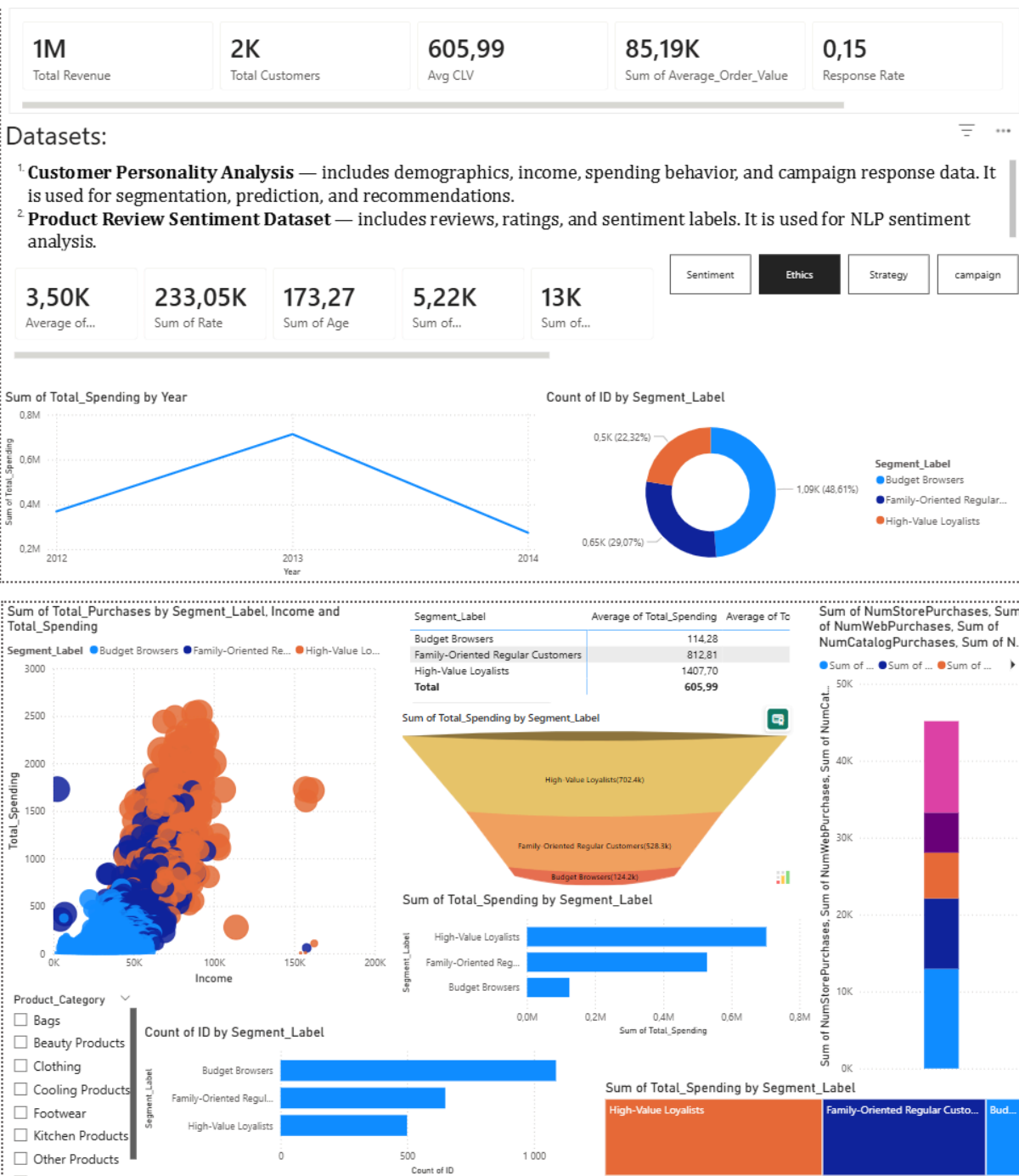


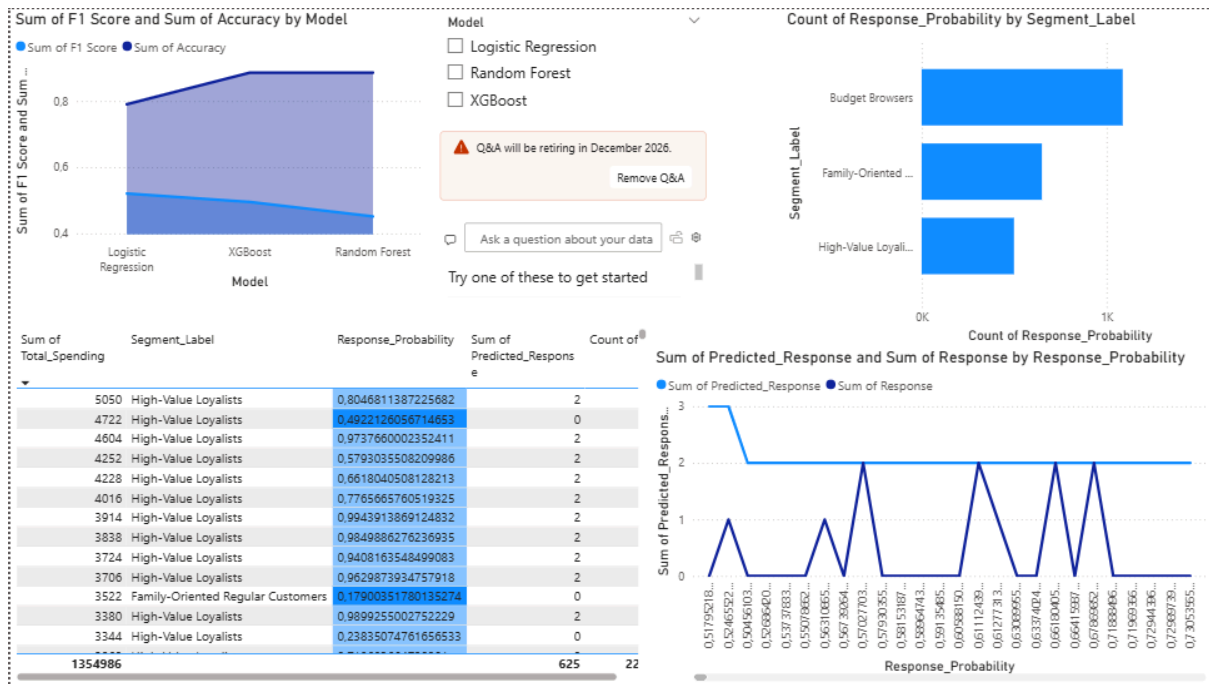


[illegible]



from power bi dashboard





2,33

Average of...

46,66

Average of Recommendation_Score

0,33

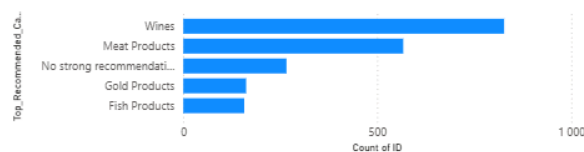
Average of Response_Probability

Segment_Label	Fish Products
Budget Browsers	87
Family-Oriented Regular Customers	36
High-Value Loyalists	34
Total	157

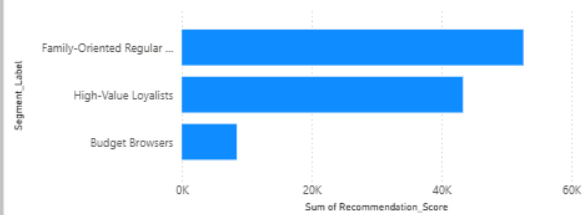
Recommendations

- Prioritize high-spending segments with personalized offers and loyalty programs.
- Target customers with high response probability to improve campaign efficiency.
- Recommend product categories based on customer behavior instead of generic campaigns.
- Use recommendations to promote categories that similar customers purchase more.
- Apply discounts and simple offers to engage low-income / low-spending customers.
- Retarget customers who responded to previous campaigns.
- Use personalized recommendations to increase average order value

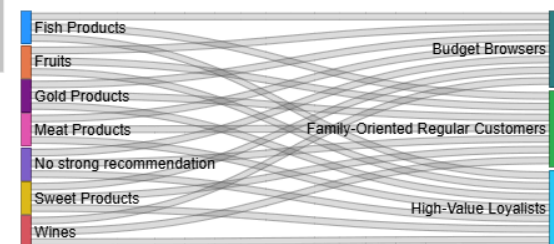
Count of ID by Top_Recommended_Category



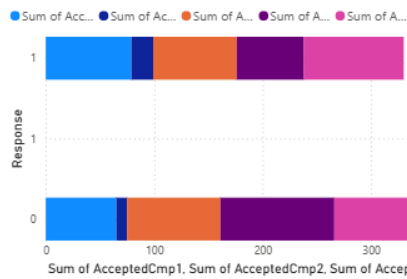
Sum of Recommendation_Score by Segment_Label



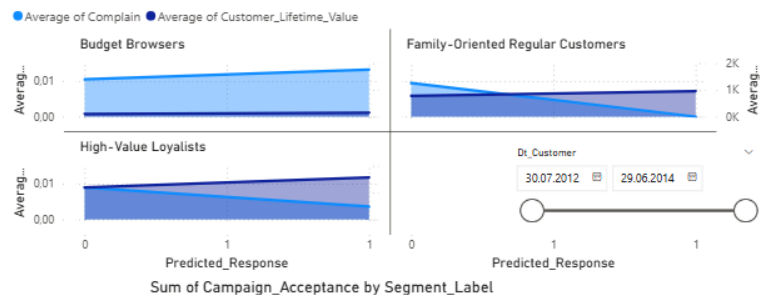
Top_Recommended_Category and Segment_Label



Sum of AcceptedCmp1, Sum of AcceptedCmp2, Sum of AcceptedCmp3, Sum of AcceptedCmp4 and Sum of AcceptedCmp5 by Response



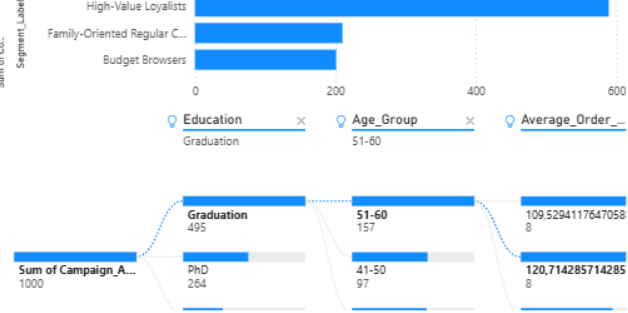
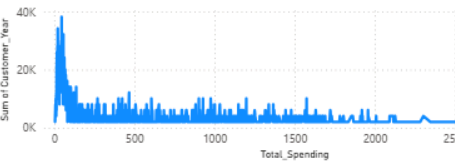
Average of Complain and Average of Customer_Lifetime_Value by Predicted_Response and Segment_Label



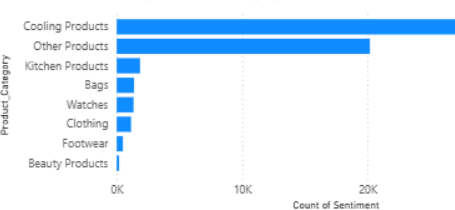
Sum of Campaign_Acceptance and Sum of Complain by Education



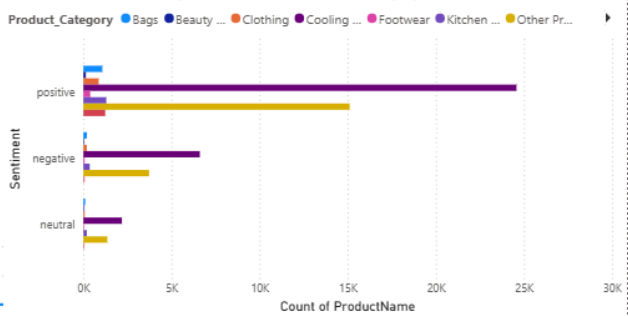
Sum of Customer_Year by Total_Spending



Count of Sentiment by Product_Category



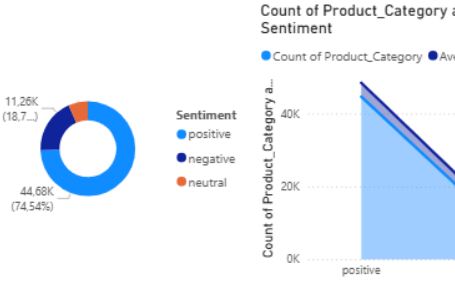
Count of ProductName by Sentiment and Product_Category



Count of Sentiment by Product_Category



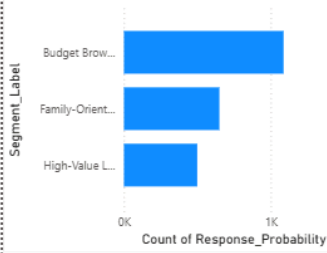
Count of Review by Sentiment



Count of Product_Category and Average of ProductPrice by Sentiment



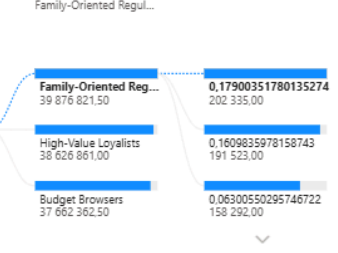
Count of Response_Probability by Segment_Label



Key influencers Top segments

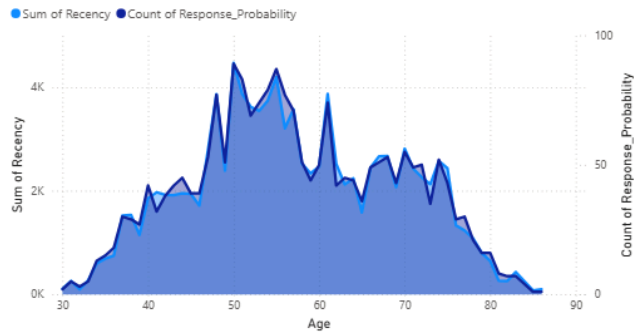


Segment_Label x Response_Proba...

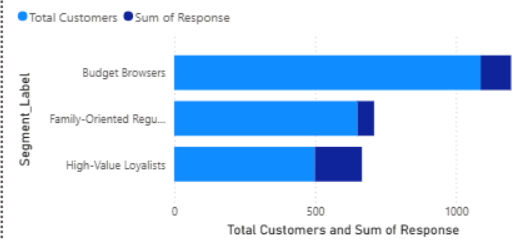


Segment_Label	Age_Group	Count of Response_Probability	Sum of Income
Budget Browsers	51-60	374	13 463 464,00
Budget Browsers	41-50	314	10 173 836,50
Family-Oriented Regular Customers	51-60	203	12 722 697,00
Family-Oriented Regular Customers	61-70	202	12 337 252,50
Budget Browsers	61-70	181	6 770 135,00
Family-Oriented Regular Customers	70+	146	8 677 098,50
High-Value Loyalists	70+	110	8 468 727,50
Budget Browsers	30-40	109	2 980 310,00
Budget Browsers	70+	109	4 274 617,00
High-Value Loyalists	61-70	109	8 402 598,00
High-Value Loyalists	51-60	108	8 398 489,50
High-Value Loyalists	41-50	98	7 724 494,50
Family-Oriented Regular Customers	41-50	93	5 765 438,00
High-Value Loyalists	30-40	74	5 632 551,50
Family-Oriented Regular Customers	30-40	6	374 335,50
Total		2236	116 166 045,00

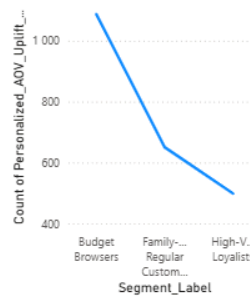
Sum of Recency and Count of Response_Probability by Age



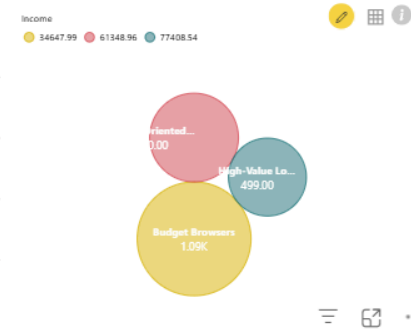
Total Customers and Sum of Response by Segment_Label



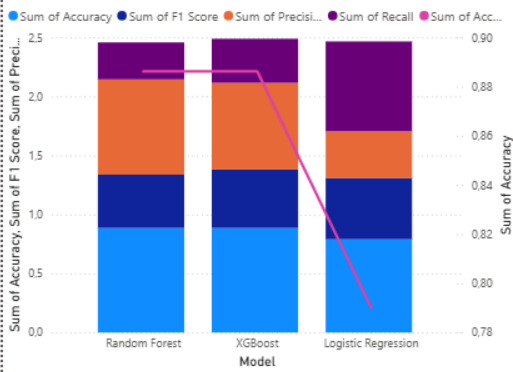
Count of Personalized_AOV_Uplift_Proxy by Segment_Label



Sum of Customer_Count by Segment_Label and Income



Sum of Accuracy, Sum of F1 Score, Sum of Precision and Sum of Recall by Model



Response_Probability	Top_Recommended_Category	Sum of Total_Spending	Segment_Label
0.8046811387225682	No strong recommendation	5050	High-Value Loyalists
0.4922126056714653	Fruits	4722	High-Value Loyalists
0.9737660002352411	Fruits	4604	High-Value Loyalists
0.5793035508209986	Sweet Products	4252	High-Value Loyalists
0.6618040508128213	Gold Products	4228	High-Value Loyalists
0.7765665760519325	Fruits	4016	High-Value Loyalists
0.9943913869124832	Gold Products	3914	High-Value Loyalists
0.9849886276236935	Fish Products	3838	High-Value Loyalists
0.9408163548499083	Gold Products	3724	High-Value Loyalists
0.9629873934757918	Wines	3706	High-Value Loyalists
0.17900351780135274	Meat Products	3522	Family-Oriented Regular Customers
0.9899255002752229	Meat Products	3380	High-Value Loyalists
0.23835074761656533	Wines	3344	High-Value Loyalists
Total		1354986	